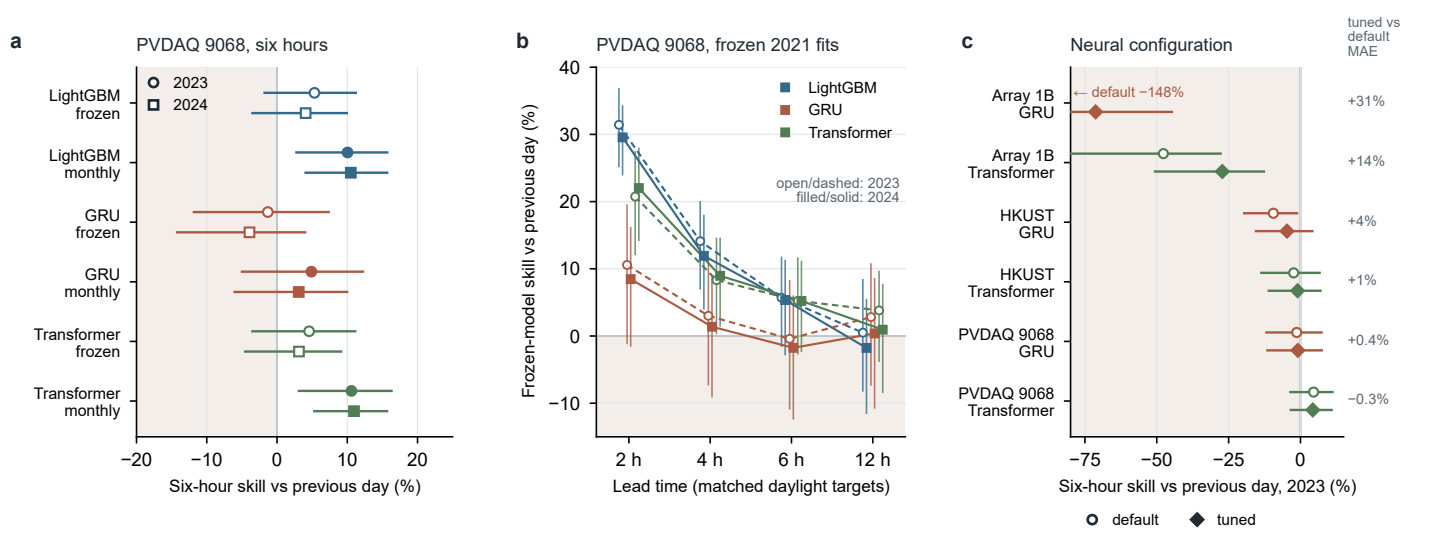
a
PVDAQ 9068, six hours
2023
2024
LightGBM frozen
LightGBM monthly
GRU frozen
GRU monthly
Transformer frozen
Transformer monthly
−20
−10
0
10
20
Six-hour skill vs previous day (%)
b
PVDAQ 9068, frozen 2021 fits
LightGBM
GRU
Transformer
open/dashed: 2023
filled/solid: 2024
Frozen-model skill vs previous day (%)
40
30
20
10
0
−10
2 h
4 h
6 h
12 h
Lead time (matched daylight targets)
c
Neural configuration
tuned vs default MAE
← default −148%
Array 1B GRU
Array 1B Transformer
HKUST GRU
HKUST Transformer
PVDAQ 9068 GRU
PVDAQ 9068 Transformer
+31%
+14%
+4%
+1%
+0.4%
−0.3%
−75
−50
−25
0
Six-hour skill vs previous day, 2023 (%)
default
tuned